

31. Morphological Operations Using OpenCV – Opening Technique

Aim

To perform morphological opening on an image using OpenCV.

Procedure

1. Import the required OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a structuring element (kernel).
5. Apply the opening operation using erosion followed by dilation.
6. Display the original and processed images.
7. Save the output image.

Program

```
import cv2

import numpy as np

# Read the input image

img = cv2.imread("input.jpg")

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create structuring element

kernel = np.ones((5, 5), np.uint8)
```

```
# Apply morphological opening
opening = cv2.morphologyEx(gray, cv2.MORPH_OPEN, kernel)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Opening Image", opening)

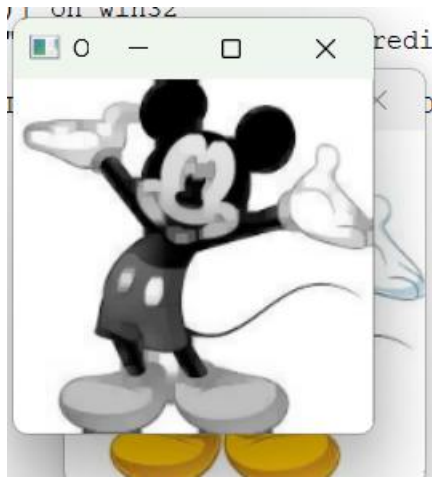
# Save output image
cv2.imwrite("opening_output.jpg", opening)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the morphological opening operation was successfully performed using OpenCV. It helps to remove small noise and unwanted objects from the image while preserving the main shape of the objects.